Diffractive Nonlocal Metasurfaces

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Metasurfaces are ushering in an era of multifunctional control over optical wavefronts realized with ultrathin planarized devices. Recent advances have been enabling unprecedented control over the frequency response of these surfaces, suggesting that the future of flat optics may tailor both spectral and spatial degrees of freedom in highly multispectral and multifunctional devices. Diffractive nonlocal metasurfaces are opening new opportunities in this direction: they leverage symmetry-protected scattering from quasi-bound states in the continuum and, by spatially manipulating controlled geometric perturbations, they support ultrasharp optical responses with wavefront-manipulating features. Encoded in nonlocal (i.e., spatially extended) resonant modes, the resulting response is observed exclusively within the bandwidth of the resulting Fano resonance, affording ideal features for a wide range of applications. In this perspective, this novel class of metasurfaces are discussed in the broader context of flat optics, highlighting their peculiar operation in contrast to relevant predecessors, and highlighting the opportunities for future advancement and applications. In particular, it is emphasized that nonlocality and selectivity are inherently related, but that spectral and spatial selectivity can be independently tuned in suitably tailored metasurfaces. In turn, this freedom allows the design and implementation of both wavefront-shaping and wavefront-selective devices. The novel optical responses, combined with the compatibility with rational design, herald new prospects for active, nonlinear and quantum metasurfaces, ultrathin devices for augmented reality, and compact tailored optical sources.

1. Introduction

In the "meta-" era of optics, a primary driver of research is to ask not "how does light behave?", but rather "how do we make light behave as we want?"[1] To this end, deliberately connecting geometrical degrees of freedom to optical degrees of freedom has become a principal goal of contemporary photonics research. The multivariate nature of electromagnetic waves solicits interest in device architectures with a high degree of multifunctionality,[2] or the ability to simultaneously manipulate several aspects of the optical response. At the same time, the desire to compactify optical devices has led the community toward the paradigm of flat optics. "Metasurfaces" are a key emerging technology tackling these challenges, capable of generating designer wavefronts through subwavelength diffractive elements.[3–11] While a monochromatic wavefront has four degrees of freedom at each location (amplitude, phase and polarization vector), a broadband wavefront has, in principle, an arbitrary number of degrees of freedom (four at each frequency and location). Therefore, recent demonstrations of impressive multifunctionality in metasurfaces[9,10,12–20] represent only a small portion of what is possible, focusing primarily on mastering the spatial degrees of freedom, irrespective of the spectral response. To move beyond these limitations, metasurfaces must be capable of simultaneously commanding both the spatial and the spectral features of the optical wavefront.

High-Q photonics, in which the quality factors (or Q-factors) are large, offer a platform capable of manipulating light with extreme spectral resolution.[21–26] However, in this context the simultaneous spatial manipulation of the optical wavefront has been largely limited. Resonant diffractive optics[27–31] is capable of distributing power between diffraction orders over narrow bandwidths, but are constrained in their operation to specific incidence angles governed by the periodicity of the device, precluding general multifunctionality or flexible custom wavefront generation. Additionally, they are mostly limited to manipulating linear polarization states. Guided mode resonances[32–39] with spatially varying reflectance have been explored,[40–44] aimed at spatially shaping the amplitude profile of resonantly scattered light, but not its phase or polarization, and not with high spatial frequencies. Recent work[45–47] incorporating guided mode resonances into metagratings[48–58] has in part overcome this limitation, but it is thus far capable only of anomalous deflection of light towards one direction relative to the metagrating lattice, while the orthogonal direction is reserved for guided mode propagation.[47] In contrast, as opined in,[47] a high-Q metasurface approach capable of arbitrary, two-dimensional spatial wavefront control represents a significant advance for the field, opening up a wide array of opportunities.

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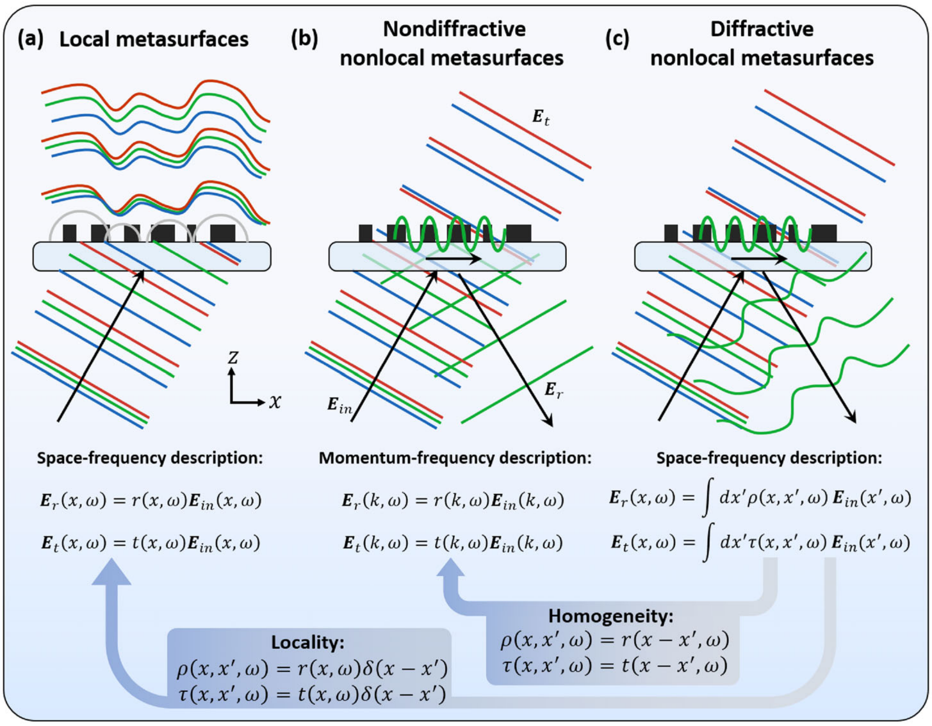

Figure 1. Comparison of three types of metasurfaces (depicted with exemplative functionalities). a) Local metasurfaces scatter incoming light based on interactions with spatially localized modes, an operation described in the space-frequency domain by the reflection and transmission coefficients $r(x, \omega)$ and $t(x, \omega)$. b) Nondiffractive nonlocal metasurfaces scatter incoming light based on interactions with spatially extended (nonlocal) modes, an operation described in the momentum-frequency domain by the reflection and transmission coefficients $r(k, \omega)$ and $t(k, \omega)$. c) Diffractive nonlocal metasurfaces scatter incoming light based on interactions with spatially extended (nonlocal) modes, an operation described in the space-frequency domain by the nonlocal reflection and transmission kernels $\rho(x, x', \omega)$ and $\tau(x, x', \omega)$. The waves depicted at the interfaces of (b,c) are guided modes mediating the nonlocal response.

In this perspective, we clarify that "diffractive nonlocal metasurfaces"[60–65] based on quasi-bound states in the continuum (q-BICs) have been enabling precisely this vision: custom 2D spatial wavefronts over arbitrarily narrow bandwidths through rational design principles deeply rooted in symmetry considerations.[66,67] We can distinguish three classes of metasurfaces: local, nondiffractive nonlocal, and diffractive nonlocal metasurfaces, where the latter is the most general category (i.e., reducing to one of the former categories in specific limiting cases), and is capable of vastly advancing the number of degrees of freedom at our command. To place these results in proper context, we differentiate nonperturbative (having low and middling Q-factors: "low-Q" and "mid-Q") and perturbative (high-Q) approaches to spatial manipulation of light, further separated into 0th order gratings, metagratings and metasurfaces. We overview key physical differences between local and nonlocal metasurfaces, including a distinction between local and nonlocal phase gradients, and highlighting their tailored selectivity to frequency, polarization, and even wavefront shape. In particular, we demonstrate that spectral and spatial selectivity are independently tunable in nonlocal devices, with the consequence that diffractive nonlocal metasurfaces may operate as wavefront-shaping devices (e.g., producing a custom quasi-monochromatic wavefront from an unpatterned broadband planewave) or as wavefront-selective devices (resonating only for a specific incoming wavefront and frequency). Finally, we discuss a number of opportunities and applications made possible by these advances, suggesting a set of fruitful directions for future research.

2. Local and Nonlocal Photonics

2.1. Metasurface Classification

A foundational design principle for metasurfaces is the assumption of locality, or the independence of the scattering response at a given location from the fields or structures at other locations. This assumption allows the straightforward design of an array of scatterers to form a metasurface, tailored to apply a wavefront transformation of choice, by reference to a library of "meta-unit" geometries whose response is assumed to not vary as the surrounding elements are modified. While only approximate, this assumption holds well for high index structures[12] and plasmonic meta-units[3] due to the strong lateral confinement of optical fields within the scatterers. Such narrowness of optical response in real space carries an associated broadness of response in reciprocal space; as a consequence, "local metasurfaces" are well known for their broadband and broad angle functionalities, as depicted in **Figure 1**a. Local metasurfaces are commonly employed to realize metalenses, vortex generators, spectrometers, and holograms.[2–11] We note that, while Figure 1a depicts as an example an idealized transmissive local metasurface, local metasurfaces are also commonly designed to be reflective.

In contrast, in a nonlocal device the scattering at a given location depends on the fields and/or structures at distant locations. This is often mediated by extended or propagating modes, such as in guided mode resonance filters, wherein the coupling to and from leaky waves supported by a corrugated dielectric slab interferes with a broad resonance to create sharp Fano resonances.[25] The broad extent of the scattering event in real space confers a narrow response in reciprocal space, manifested as spectral and angular selectivity. In other words, the resonant response is observed only for frequencies and momenta closely matching the dispersion of the leaky mode. This dispersion is governed by the band structure of the infinite, periodic device, whose period is commonly subwavelength (a so-called "0th order grating") in order to guarantee complete interference and unity reflectance in the absence of loss,[68] and resulting in a response limited to specular reflection and transmission [see Figure 1b]. Such nondiffractive nonlocal devices can operate as bandpass filters,[69–71] refractive index sensors,[72–75] nonlinear optical elements,[76–79] active photonic elements,[80,81] and image differentiators.[82–85]

In the general case, a nonlocal mode whose coupling to external waves is spatially customizable can combine both the spatial control typical of local metasurfaces [Figure 1a] and the frequency selectivity of nonlocal metasurfaces [Figure 1b]. Such diffractive nonlocal devices [Figure 1c] can be thus seen as the most general category of metasurfaces. Since the assumption of locality is only ever approximate, local metasurfaces may be conceptualized as the limiting case of diffractive nonlocal metasurfaces when the nonlocality and associated selectivity is very low. Nondiffractive nonlocal metasurfaces, on the other hand, are the limiting case when subwavelength periodicity is enforced, removing all diffraction orders other than the 0th order (the specular response).

It is instructive to recover the simplified form of the scattering in Figure 1a,b by applying appropriate approximations to the generalized form in Figure 1c. Taking the reflection equations as an example (the transmission equations are analogous), we begin with the general form that applies to all three classes in Figure 1

$$\mathbf{E}_r(x,\omega) = \int dx' \rho\left(x,x',\omega\right) \mathbf{E}_{in}\left(x',\omega\right) \tag{1}$$

In the case of local metasurfaces, by definition the nearest neighbors do not impact the scattering. In other words, the scattering kernel ρ is spatially instantaneous, and takes the form

$$\rho\left(x,x',\omega\right) = r(x,\omega)\,\delta\left(x-x'\right) \tag{2}$$

where δ is the Dirac Delta function. Insertion into (1) and integrating yields

$$\mathbf{E}_r(x,\omega) = r(x,\omega)\,\mathbf{E}_{in}(x,\omega) \tag{3}$$

This form is consistent with the foundational vision of metasurfaces, as elements providing point-by-point manipulation of the scattered wavefront.

In the case of nondiffractive nonlocal metasurfaces, we cannot make the assumption that the response is spatially instantaneous, but instead treat the surface as homogenized, that is, that the discrete subwavelength periodicity of the structure may be approximated as continuous translation symmetry. Mathematically, ρ may be written in the shift-invariant form

$$\rho\left(x,x',\omega\right) = r\left(x-x',\omega\right) \tag{4}$$

Then, by the convolution theorem, insertion of Equation (4) into Equation (1) and Fourier transformation of the result gives

$$\mathbf{E}_r(k,\omega) = r(k,\omega)\,\mathbf{E}_{in}(k,\omega) \tag{5}$$

In this sense, diffractive nonlocal metasurfaces are the most general class depicted in Figure 1, abiding no such simplified scattering equation. This added complexity in comparison to the common simplified approaches confers unique opportunities to powerfully command both the spatial and spectral properties of scattered light.

2.2. Metasurfaces with Varying Degree of Nonlocality

While Figure 1 depicts a classification cleanly separating metasurfaces into "local" and "nonlocal" categories, in truth no metasurface is fully "local", and instead all metasurface exist on some continuum of nonlocality. Famously, even the bare interface of a blackbody radiates light with nonzero spatial coherence,[86] owing to the fact that the Fresnel coefficients of an interface necessarily change with the angle, approaching unity at arbitrarily grazing angles of incidence (this variation in reciprocal space confers a finite extent in real space). **Figure 2** qualitatively depicts such a continuum, with example metasurface categories placed relative to each other according to their typical operation.

Toward one extreme, the characteristic size of the scattering is subwavelength. For instance, isotropic effective media (Figure 2a) composed of deeply subwavelength inclusions represent a class of metamaterials (or metasurfaces when made thin) that are as local as possible (effectively an unpatterned interface). Local metasurfaces composed of plasmonic antennas (Figure 2b) may be composed of deeply subwavelength unit cells,[3] implying highly localized responses. Dielectric metasurfaces composed of pillars with moderate to high refractive index (Figure 2c) are typically less localized than their plasmonic counterparts, but still abide the local approximation quite successfully. However, recently efforts to engineer the angular dependence of dielectric metasurfaces have extended this category to the partially nonlocal regime.[87]

Near the "gray" area of the continuum (the transition point between "local" and "nonlocal"), the response is localized at a scale near the wavelength of light. On the local side of this transition are bianisotropic metasurfaces,[88] seen in Figure 2d, distinguished by their support of magnetoelectric coupling, that is, the excitation of magnetic (electric) polarizations in the near field due to electric (magnetic) incident polarizations. This coupling leads to a nonlocal polarization response, and bianisotropic metamaterials enable interesting opportunities in asymmetric transmission[90–93] and reflection,[94,95] anomalous refraction,[96–99] and polarization manipulation including optical activity.[100–104] Relatedly, Huygens' metasurfaces,[105–108] seen in Figure 2e, are composed of structures supporting overlapping magnetic and electric resonances, and are well known to depend

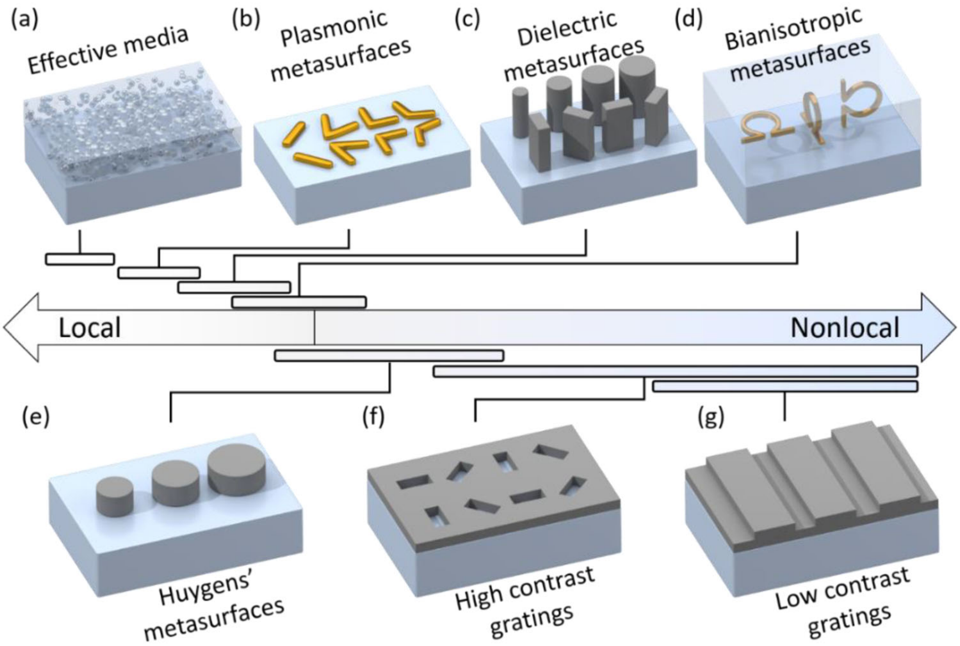

Figure 2. Class of metasurfaces qualitatively arranged by typical degree of nonlocality. a) Effective media composed of deeply subwavelength inclusions may approximate unpatterned thin films. b) Plasmonic metasurfaces may be composed of subwavelength antenna, supporting highly local responses. c) Dielectric metasurfaces are less localized than their plasmonic counterparts. d) Bianisotropic metasurfaces control the electric and magnetic response, yielding an angle-dependent response with corresponding nonlocality. e) Huygens' metasurfaces overlap two angle-dependent resonances to control light with strong forward scattering. f) High contrast gratings, including diffractive nonlocal metasurfaces, are composed of deeply corrugated high index systems based on periodic or highly regular lattices. They represent a wide range of nonlocality, engineerable by band structure and Q-factor engineering. g) Low contrast gratings are less localized than their high contrast counterparts, composed of shallow corrugations.

strongly on the angle of incidence, often an undesirable manifestation of nonlocality in this case.[109]

Finally, toward the nonlocal extreme of the continuum are devices supporting long-lived states that convey energy for long distance in the plane of the device. In high contrast gratings, seen in Figure 2f, deeply corrugated structures composed of high index materials may support a wide range of localization degree, all the way from devices requiring only a few grating periods to ultra-high Q-factor devices requiring many gratings periods.[89] Furthermore, band structure engineering may be used to explicitly control the degree of localization from a Dirac cone to a flat band[110] and may be tuned independently of the Q-factor.[111] In this context, we may consider high contrast gratings as the most flexible category of device, and indeed is the category in which the diffractive nonlocal metasurfaces (the primary focus of this work) are best placed. In low contrast gratings (Figure 2g) or guided mode resonance filters,[32–36] the Bragg scattering and related effects that confer localization in high contrast gratings are absent, and modes approaching truly "global" responses may be supported in principle. In particular, the characteristic distance over which the modes carry optical energy follows roughly $Q\lambda_0/n_{eff}$, where Q is the Q-factor, λ_0 is the free-space wavelength, and n_{eff} is the effective index of the guided mode; as Q tends towards infinity, so too does the nonlocality.

3. Recent Advances in Nonlocal Flat Optics

Here we overview recent efforts on engineering nonlocality in flat optical devices. First, we distinguish nonperturbative metastructures (low-Q and mid-Q) from perturbative metasurfaces (high-Q). Within each of these sub-classes we can also distinguish between 0th order gratings, metagratings, and metasurfaces. While the focus of this perspective is on high-Q metasurfaces capable of manipulating wavefronts, we briefly describe the other categories to put these recent advances into a broader context.

We refer to "0th order gratings" as devices characterized by a subwavelength periodicity such that only specular reflection and transmission, that is, 0th order diffraction, are supported within the operative incident conditions. In these devices, the period Λ is smaller than the wavelength λ, that is, $\Lambda/\lambda < 1$, and the incident angle is chosen such that the far-field response is nondiffractive, that is, it does not support anomalous diffraction or wavefront shaping. In "metagratings", the meta-unit cells are larger than the operating wavelength, and the geometry of these unit cells is designed to manipulate the higher diffraction orders (in amplitude, phase and polarization). The meta-units may then be arranged regularly or in a spatially varying manner.[51–55] In this case, $1 < \Lambda/\lambda < N$ where N is on the order of 3 (there is no precise cutoff). In the "metasurface" approach, in contrast, meta-unit cells are themselves subwavelength, but the geometry of the final metasurface is configured based on arrays of these meta-units, such as into phase gradients with an overall periodicity $\Lambda/\lambda > N$. We note that the term "metasurface" is used quite broadly in the literature, often encompassing devices in any of the three categories above. For instance, we may alternatively refer to 0th order gratings as "nondiffractive nonlocal metasurfaces", as in Figure 1.

An example of each category is shown in **Figure 3**, where along the horizontal dimension we show examples varying Λ/λ, and

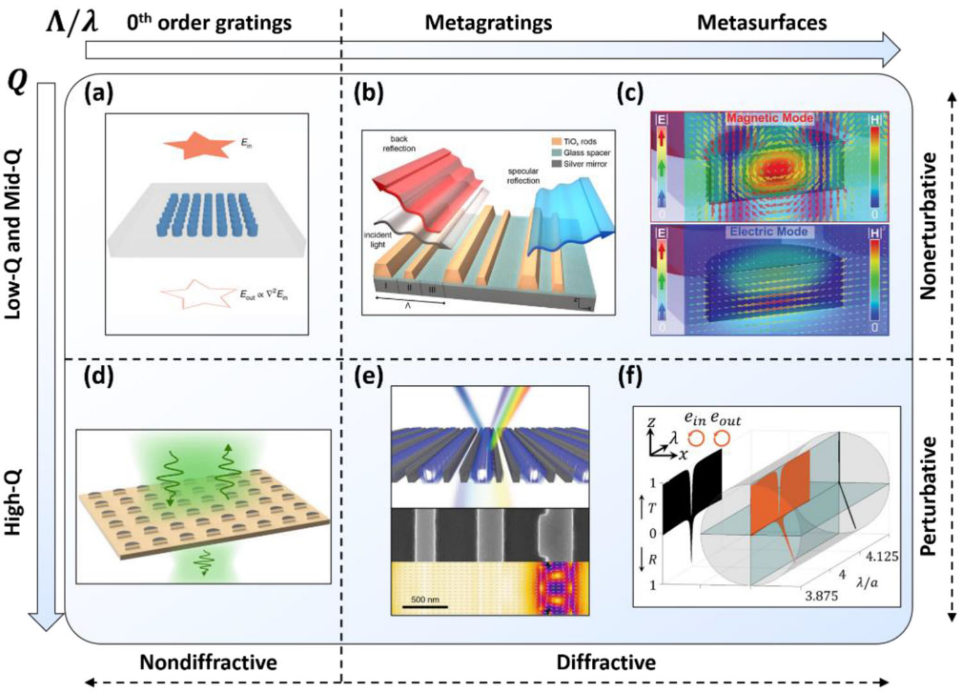

Figure 3. Nonlocal meta-structures. a) Flat optics for image differentiation is an example of 0-th order gratings with low or moderate Q-factors. Reproduced with permission.[85] Copyright 2020, Springer Nature. b) Metagratings have been demonstrated to support broadband anomalous reflection features. Reproduced with permission.[50] Copyright 2017, American Chemical Society. c) Huygens' metasurfaces rely on overlapping magnetic and electric resonances for operation with high transmission efficiency. Reproduced with permission.[107] Copyright 2015, Wiley. d) Q-BICs are commonly studied in nondiffractive devices for their unbounded Q-factors controlled through a symmetry-breaking perturbation. Reproduced with permission.[24] Copyright 2018, American Physical Society. e) Metagratings with a symmetry-breaking perturbation support high-Q features that also control the efficiency of higher diffraction orders. Reproduced with permission.[46] Copyright 2020, Springer Nature. f) Diffractive nonlocal metasurfaces are capable of arbitrary anomalous diffraction over arbitrarily narrow bandwidths. Reproduced with permission.[62] Copyright 2021, American Physical Society.

along the vertical dimension we differentiate by Q-factor. The vertical dimension may alternatively be viewed as differentiating between nonperturbative (Figure 3a–c) and perturbative approaches (Figure 3d—f), the latter having rational control of Q over an unbounded range through symmetry-breaking.[24] We emphasize that, while nondiffractive flat optics of this latter category (Figure 3d) have been studied for decades, their diffractive counterparts (Figure 3e,f) have only been introduced recently. In our view, the high-Q diffractive nonlocal metasurfaces (Figure 3f) are a particularly promising avenue for future research into comprehensively commanding the spatial and spectral properties of light (see also ref. [47]).

3.1. Nonperturbative, Nonlocal Flat Optics

3.1.1. Nonperturbative 0th Order Gratings

In a nonperturbative 0th order grating, the unit cells are designed to perform a desired operation on the polarization, broad spectral, or momentum properties. We highlight two examples of 0th order gratings employing nonlocal modes for the control of the momentum of light waves exiting the device, aiming at applying linear mathematical operations to the impinging wavefront.[82] First, the nonlocality of periodic structures was employed to realize image differentiation, leveraging the fact that the optical transfer function in momentum space may be engineered to mimic spatial derivatives.[83,84] This functionality was then realized for two-dimensional image differentiation in simple-to-fabricate dielectric structures operating in the optical domain,[85] as shown in Figure 3a. The transfer function may also be controlled to emulate the propagation kernel of free space, with the effect of "squeezing" the free-space between optical elements.[112–114] This advance suggests that by using metasurfaces, optical systems may be compactified past the degree of compactification afforded by simply thinning the optical elements themselves.

3.1.2. Nonperturbative Metagratings

Low-Q, or nonperturbative metagratings (Figure 3b) have received significant attention lately, owing to their compatibility with high efficiency operation at large deflection angles. After it was clarified that local phase manipulation is insufficient for unitary efficiency beam steering,[48] the concept of metagratings was introduced,[49] achieving unitary efficiency by leveraging nonlocality. These concepts were then realized experimentally[50] and extended to transmissive-type devices.[51] Interestingly, by arraying distinct metagrating unit cells across a device, spatial manipulation of the nonzero diffraction orders may be obtained.[52–55] Many inverse design approaches fall into this category,[56–58] yielding high efficiency anomalous deflection by optimizing unit cells with periodicities $\frac{\Lambda}{\lambda} > 1$. This approach greatly exceeds the capabilities of traditional metasurface approaches for large angle deflections. Generally, however, while some control is possible as the number of diffraction orders grows,[59] the approach becomes increasingly complicated for the larger periods necessary for low-angle deflections. In this case, the metasurface approach is still commonly preferred.

3.1.3. Nonperturbative Metasurfaces

Broadly construed, the category of nonperturbative metasurfaces incorporates the majority of recent literature on metasurfaces. However, the vast majority of these operate assuming locality. Here, we highlight Huygens' metasurfaces[105–108] that are often composed of thin, dielectric structures supporting both magnetic and electric Mie resonances,[107] seen in Figure 3c. When these two resonances are spectrally overlapped, they exhibit a resonant crossing due to their opposite decay symmetry (see also ref. [115]), supporting high transmission across the resonances, with a phase varying over 2π. By adjusting the size and shape of the unit cells, the resonances may be spectrally shifted, accessing any phase delay value. In this way, such devices can apply a desired wavefront transformation by locally adjusting the shape according to the phase, best exemplified by metasurface holography.[116]

However, despite being designed and conceived as local metasurfaces, the performance of dielectric Huygens' metasurfaces has recently been demonstrated to suffer due to the inherent nonlocality of the resonant modes responsible for their function.[109] The resonances are supported by several adjacent structures in the periodic array, and have meaningful resonant frequency dispersion over moderate changes in incident angles. Moreover, the dispersion may differ between the two "accidentally" aligned electric and magnetic type modes, which leads to a destruction of the spectral overlap and therefore of the high transmission condition at sufficiently oblique angles. Since by reciprocity a phase gradient Huygens' metasurface designed to anomalously deflect normally incident light must identically deflect the oblique, time-reversed wave to normal incidence, and since this oblique wave is known to not properly excite the two involved modes, the anomalous transmission efficiency must also suffer.[109] We emphasize these limitations of Huygens' metasurfaces in contrast to the perturbative diffractive nonlocal metasurface approach detailed below, which does not suffer from these challenges despite the increase in both Q-factor and degree of nonlocality. In essence, the key difference is that in this latter approach the geometric phase is used rather than the spectral phase, enabling us to engage a single band of modes with external light independently of the incident angle.

3.2. Perturbative Nonlocal Flat Optics

3.2.1. Perturbative 0th Order Gratings

In a perturbative 0th order grating, a symmetry-lowering perturbation is applied to an otherwise high symmetry array of subwavelength optical structures. While many examples of this approach have been reported in the past few decades (see ref. [117] as a notable example), interest was recently reignited in this area when the unifying concepts behind quasi-bound states in the continuum (q-BICs) were clarified.[24] A q-BIC (Figure 3d) is a mode that leaks to free-space only in the presence of a perturbation, and it is otherwise bound to the scatterer by symmetry protection, even if it has momentum compatible with the radiation continuum. Since, generally, the radiative lifetime is inversely proportional to the scattering strength squared (see, for instance, ref. [118]), q-BICs obey a universal dependence $Q \propto 1/\delta^2$ for sufficiently small δ, where δ is a characteristic magnitude of the geometric perturbation.[24] In the absence of a geometric perturbation, the mode is a bound state in the continuum (BIC).[21–23] As an alternative to geometric perturbation, changing the incident angle may break the relevant symmetry[119–121] and hence the incident in-plane momentum k plays the role of δ, yielding $Q \propto 1/k^2$ for sufficiently small angles. The same essential physics is manifested in guided mode resonance (GMR) filters,[33] which are periodically corrugated slabs; here, symmetry breaking consists in breaking the continuous translation symmetry, and δ is often the depth of corrugation.

However, a conceptual distinction between GMR filters and q-BIC devices is that in GMR filters the modes are not in the continuum in the absence of the perturbation; rather, the perturbation brings the modes into the continuum by breaking translation symmetries, while also introducing a net dipole moment by breaking the relevant spatial symmetries. Another functional distinction is that such low contrast structures provide little to no control over the spatial localization of energy; the size of array must scale with the Q-factor, requiring large devices for high-Q operation.[69] In comparison, q-BICs in high contrast structures may be relatively highly localized states,[89,122] affording the convenient command of the Q-factor independently of the band structure, and therefore enabling compact optical devices with both spatial and temporal confinement of light.[110,111]

Notably, in high contrast structures wherein the perturbation alters the periodicity of the device (e.g., by doubling the period), the modes are brought into the continuum (Brillouin zone folding) as well as made leaky by the perturbation.[111] Hence, both translational and reflection symmetries are broken by the perturbation, similar to GMR filters. In the absence of the perturbation, the modes are bound due to being under the light line, and are only within the continuum with respect to an artificial unit cell. For this reason, these modes have been dubbed artificial BICs,[123] suggesting the term artificial q-BICs for their leaky counterparts. We may also simply refer to both q-BIC categories by the same term, where in the case without Brillouin zone folding the emphasis is "(quasi)-(bound states in the continuum)" while in the case with Brillouin zone folding the emphasis is "(quasi-bound) (state in the continuum)". Importantly, artificial q-BICs are characterized by a Q-factor that is independent of the incident angle[111] (characteristic of a dipolar scattering profile), in contrast to the $1/k^2$ dependence noted above for states already in the continuum (characteristic of a higher order multipole[124]). This dipolar scattering profile has the meaningful consequence that the q-BIC engages light equally for any incident angle (i.e., across the band of modes), suggesting compatibility with the Huygens' principle and hence phase-front manipulation through spatial tuning of the effective dipole moment.

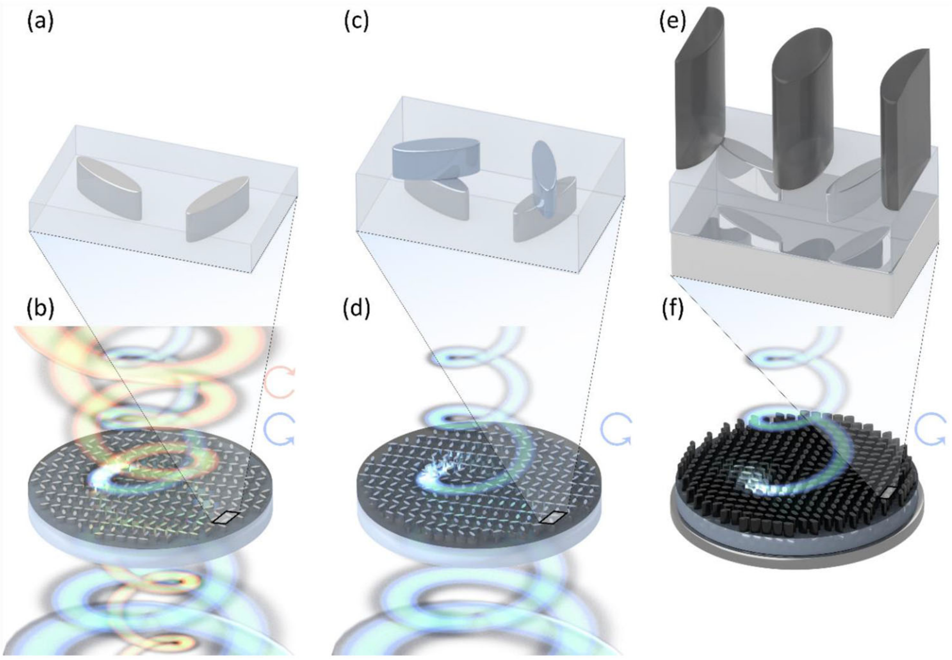

Figure 4. Diffractive nonlocal metasurfaces encoded with spatially varying wavefronts. a) Achiral unit cell used in Refs. [60, 61]. b) Schematic depicting converging/diverging OAM beams encoded into a q-BIC based on the achiral units in (a). For light scattered upwards, circular polarization of each handedness encodes a converging (diverging) OAM with positive (negative) integer spin. c) Chiral unit cell used in refs. [62, 63] d) Same as (b) but with the units in (c), eliminating one polarization. e) Chiral and unidirectional unit cell used in ref. [64] f) Same as (d) but with a back mirror, eliminating a hemisphere.

Beyond their defining scalar property governing the Q-factor, q-BICs are fundamentally vectorial objects, obeying selection rules that govern their polarization dependence.[60] Ref. [60] exhaustively catalogued the selection rules for both monoatomic and multi-atomic (i.e., zone-folded) lattices, while ref. [61] showed that different combinations of these perturbations may be added successively to the lattice to yield independent degrees of control over the optical response. These knobs are tunable independently of each other to first order in perturbation theory, and hence q-BICs are an intrinsically highly multifunctional phenomena (as an example, up to eight scattering parameters may be controlled simultaneously). Finally, by stacking perturbations vertically, breaking out-of-plane symmetries to yield a generally chiral response,[125–127] ref. [62] extended the selection rules to the entire Poincaré sphere within a rational design scheme. In stark contrast to conventional Fano resonances, the response to circular polarization in these devices exhibits unitary reflectance and full circular dichroism, with a resonant phase that is arbitrarily tunable by geometric phase engineering. Crucially, this phase is independent of the resonant frequency, that is, it is distinct from the spectral phase profile.

3.2.2. Perturbative Metagratings

While in certain cases the bandwidth of the metagrating approach may become quite narrow,[49] recent work incorporated symmetry-controlled Q-factors on the order of 10^3 into the metagrating architecture.[46] In this approach (Figure 3e), one-dimensional silicon fingers populate a superwavelength unit cell that supports several diffraction orders. In the direction parallel to the fingers, subwavelength corrugations (or perturbations to an otherwise continuous translational symmetry) are introduced, coupling external light to guided modes highly reminiscent of waveguide modes well established in silicon photonics. A sharp Fano resonance is observed, born of the interference of the coupling to this mode and Fabry–Perot resonances. Due to the superwavelength periodicity, light may diffract to higher-order channels upon transmission or reflection. By judiciously adjusting the size of the non-corrugated fingers, a degree of control over these diffraction efficiencies is obtained, leading to metagratings that anomalous deflect to a desired oblique angle over a narrow bandwidth of choice. However, since along the fingers the periodicity is subwavelength, this functionality is fundamentally limited to one direction.[47]

3.2.3. Perturbative Nonlocal Metasurfaces

Finally, the most recent class of devices in Figure 3, capable of arbitrary anomalous diffraction with unitary efficiency and exclusively within arbitrarily sharp bandwidths (Figure 3f), consists in perturbative nonlocal metasurfaces. We review the recent developments starting from ref. [60], which derived selection rules for q-BICs, and so far culminating in ref. [64], which demonstrated the potential of q-BICs for controlling at the same time both the local and nonlocal features of light for use in custom thermal sources. Typical geometries and functionalities of these works are shown in **Figure 4**.

Selection Rules for q-BICS: While ref. [24] reinvigorated the interest in broken symmetries for flat optics by clarifying that a wide range of devices have a universal scalar property—a perturbatively controlled Q-factor—ref. [60] broadened the picture by establishing a strategy to control the vectorial features of these modes. A "selection rule" is a constraint governing whether the excitation of a state is forbidden or allowed by symmetry; since a q-BIC is by definition a state made excitable due to perturbations breaking a symmetry class, selection rules are a fundamental defining feature of these physics. Using the language of group theory, ref. [60] derived the selection rules defining q-BICs according to the space group of the resulting lattices, including the determination of which (if any) polarization may excite each mode in a given lattice. Since the behavior depends only on the space group (or set of symmetries) retained after perturbation, the precise geometrical motif of the lattice is unimportant, and there is a finite list of unique perturbations applied to a given 2D lattice. An exhaustive catalogue of selection rules is provided in ref. [60] for six simple lattice families, providing a reference and roadmap for designing both the scalar (Q-factor) and vectorial (polarization) properties of q-BICs within a perturbative framework.

Of particular interest is the flexibility of the p2 space group, which is compatible with q-BICs supporting a linear polarization state oriented arbitrarily with respect to the lattice (Figure 4a). When excited by arbitrary polarization states, the selected linear polarization component is reflected by the surface, while the orthogonal component is transmitted. When excited with circular polarization, as we vary across the metasurface aperture the orientation of the selected linear polarization, a geometric phase is introduced, for which one of the circular polarization components is transmitted or reflected with an arbitrarily varying phase profile. Since this dichroic geometric phase is introduced by a small perturbation to an otherwise periodic array of highly symmetric unit cells, the perturbation may be varied spatially without altering the predominant, unperturbed features of the lattice that govern the supported modes, implying opportunities for phase gradient functionalities exclusively incorporated to the resonant response.

Multifunctional Nonlocal Metasurfaces: Together, Refs. [60, 61] demonstrated that the addition of a phase gradient in the resonant response of nonlocal metasurfaces may be applied to both in-plane directions. The phase gradient can be applied to demonstrate focusing or other nonperiodic functionalities. Hence, these q-BICs can manipulate the spatial, polarization and spectral degrees of freedom of optical wavefronts. Depicted in Figure 4b, the q-BIC may be engineered to leak to a desired wavefront for one handedness of CP light, while the opposite handedness encodes the conjugate wave as usual for geometric phase. In this case, a converging orbital angular momentum (OAM) beam is encoded for right-hand circularly polarized (RCP) light when travelling upwards. Correspondingly, this beam is diverging when travelling downwards, and left-hand circularly polarized (LCP) light has the equal and opposite OAM and converges (diverges) while travelling downward (upward). The achirality constrains the q-BIC to engage with linearly polarized light everywhere across the device, but a spatially varying linear polarization may be decomposed into circularly polarized components that carry equal and opposite geometric phases.

Most generally, by combining two distinct perturbations (each of which, if applied in isolation, yields a distinct space group), a low-symmetry lattice can be obtained wherein each perturbation independently controls a different degree of freedom in the optical response. This includes the ability to control: i) the excitation of two independent modes of distinct symmetries, ii) the Q-factor and polarization angle of a single mode, and most generally iii) the Q-factor and polarization angles of several (up to four) modes simultaneously.[61] On this basis, it was demonstrated that, by applying three distinct spatially varying perturbations to a photonic crystal slab, three distinct anomalous diffraction angles may be imparted to three wavelengths simultaneously. This degree of multifunctionality is made possible by the perturbative nature of the underlying phenomena: tuning a geometric knob leaves the optical responses not targeted by that knob unchanged to first order in perturbation theory.

Chiral q-BICS: While an optical Fano resonance may support a geometric phase for a subset of the resonantly scattered components, the efficiency of this operation is fundamentally limited to 25% in the case of achiral devices, as those considered in refs. [60, 61]. In contrast, ref. [62] demonstrated that, by combining two different p2 space groups separated vertically (out-of-plane), the selection rules can be generalized from any linear polarization (a single degree of freedom controlled by the in-plane orientation angle of the perturbation) to arbitrary elliptical polarization (two degrees of freedom controlled by the in-plane orientation angles of each layer). The perturbation (Figure 4c) yielding such intrinsically chiral Fano resonant metasurfaces thereby controls three degrees of freedom per mode: the Q-factor and the vectorial polarization state. In contrast to the conventional Fano resonant spectral phase profile, which is fixed at the resonant frequency to π, once a circular polarization is selected, the resonantly scattered light is reflected with arbitrary resonant phase (a geometric phase).[62] This phase is hence tunable while ensuring unitary reflected amplitude (full circular dichroism). On this basis, ref. [62] extended the applicability of wavefront-shaping q-BIC physics to include devices with 100% efficiency and full chirality. In particular, the (LCP) component that is unavoidable in the achiral case (Figure 4b) may be completely eliminated in the chiral case (Figure 4d). In parallel, refs. [126, 127] demonstrated full chirality in a perturbative scheme using similar out-of-plane perturbations. However, in these works the selected circular polarization was not orientable with respect to the lattice of the device. In contrast, ref. [62] presented devices with arbitrary local phase of the circular polarization scattered from the q-BIC; it is precisely this local degree of freedom (the geometric phase) that enables wavefront-manipulation with unitary efficiency.

Wavefront-Selective Fano Resonances: Nonlocal metasurfaces can also be leveraged to enable spatial selectivity.[63] This feature can be achieved by controlling the optical response of perturbative metasurfaces with aperiodic spatial transformations, establishing that a spatially varying perturbation manipulates not only how an incident wavefront is scattered, by which incident wavefronts may be efficiently scattered. In conventional periodic gratings, a Fano resonance is excited only for specific combinations of frequency (spectral selectivity) and incident momentum (angular selectivity) matching the band structure of the underlying mode. In other words, Fano resonances have long been known to be selective to certain plane waves (and linear polarizations). In contrast, in an aperiodic device based on chiral q-BICs, the resulting Fano resonances can be made selective to a custom, arbitrary wavefront (and circular polarization) determined by the spatial profile of the applied perturbation.[63] At the band-edge frequency, the selected wavefront is phase conjugated by the wavefront transformation encoded in the perturbation. In other words, the selected wavefront is retroreflected everywhere across the device. Since the wavefront is preserved upon reflection in this way, we can consider it the "eigen-wave" of the metasurface. In Figure 4d for example, the encoded eigenwaves is the OAM beam that converges when travelling upward, or equivalently, the corresponding time-reversed beam that diverges when travelling downward.

Light incident with deviations from this eigen-wave are reflected with a smaller efficiency, negligibly interacting with the surface for sufficiently large deviations. Hence, Fano resonant metasurfaces can be generally made spatially selective, and q-BIC physics provides a rational design paradigm capable of specifying the selected wavefront. Notably, the degree of spatial selectivity can be controlled through the band structure of the underlying q-BIC: flat bands are less selective than sharp bands, meaning that band structure engineering[110] may tune the degree of spatial selectivity. Hence, in combination with the perturbative control of the Q-factor (i.e., the spectral selectivity), these results clarify that the q-BIC platform offer complete customization of the spatial and spectral selectivity of flat optical devices.

Thermal Metasurfaces: By adding optical loss to the wavefront-selective devices demonstrated in ref. [63], ref. [64] demonstrated devices exhibiting wavefront-selective absorption and thermal emission. By tuning the nonradiative lifetime to be equal to the radiative lifetime, unity absorption is possible exclusively at resonance (known as critical coupling)[128] and exclusively for the selected eigen-wave. By the modal radiation laws,[129] these devices equivalently support wavefront-selective thermal emission. When driven by heat, the critically coupled narrowband response of the resonance efficiently produces quasi-monochromatic radiation while broadband emission is negligible. While much effort has gone into commanding thermal emission in recent decades, ref. [64] introduces a "thermal metasurface" platform capable of generating quasi-monochromatic, custom wavefronts, controlling linewidth, polarization, coherence, and wavefront of thermal emission. For instance, it is possible to envision unidirectional emission of a desired circular polarization, focused thermal emission or net orbital angular momentum emission.[64]

These thermal metasurfaces typically benefit from a mirror plane to direct all emission to a single hemisphere, and a hybrid local-nonlocal phase gradient selecting polarization handedness and the angle of emission. These are controlled simultaneously with a tall monatomic pillar in the unit cell (for the local response) and a short diatomic set of pillars in the unit cell (for the nonlocal response), as depicted in Figure 4e. The local response serves the role of the bright mode or continuum of states in the Fano resonance,[25] while the q-BIC serves the role of the dark mode or discrete state. In refs. [60–63], the local response is simply transparent, imparting no transformation to the incident light over a broad bandwidth near the target operating wavelength. However, in order to satisfy the constraints due to reciprocity and time-reversal invariance in such a single-port device, it was found that the local response of a thermal metasurface must be manipulated simultaneously with the nonlocal response. In other words, in the absence of the perturbation governing the q-BIC, the device is simply a conventional, low-loss geometric phase (local) metasurface that preserves the circular polarization and anomalously reflects it over a broad bandwidth. In the presence of the perturbation, the broadband wavefront-shaping response, controlled through conventional local metasurface design principles, is punctuated by a narrowband wavefront-selective absorption (thermal emission) (Figure 4f), which is controlled by the nonlocal metasurface design principles outlined in refs. [60–63].

Unidirectional[130] and focused emission[131] has also been demonstrated in photoluminescent metasurfaces. In this approach, a quantum-well metasurface supports a surface wave near one interface of an otherwise conventional phase varying metasurface. This laterally propagating nonlocal mode scatters off the interface, and the portion that travels through the metasurface is locally shaped as it propagates vertically. We anticipate that inclusion of the q-BIC approach in ref. [64] with the experimental platform demonstrated in refs. [130, 131] may further expand the command of photoluminescence while optimizing the light extraction at the desired wavelength of operation.

4. New Phenomena in Diffractive Nonlocal Metasurfaces

In order to clarify the opportunities afforded by recent advances in diffractive nonlocal metasurfaces, here we highlight a few key and novel aspects of the phenomena underlying their operation. In particular, we emphasize two points: i) local and nonlocal phase gradients are notably distinct and ii) nonlocality confers selectivity to the optical response, and yet spatial and spectral selectivity are separately tunable. The first point clarifies that the nonlocal phase gradients reported in refs. [60–65] are incompatible with vertically symmetric thin films if unitary efficiency operation is desired, implying that the design approach in ref. [62, 63] is, in a design sense, as simple as possible. The second point distinguishes wavefront-shaping devices, which anomalously diffract incoming light over a wide range of incident wavefronts, and wavefront-selective devices, which only engage a narrow range of incident wavefronts.

4.1. Local versus Nonlocal Phase Gradients

In this section, we emphasize two primary differences between local and nonlocal phase gradients: i) a local phase gradient impacts directly to the output of the metasurface, while a nonlocal phase gradient applies to the coupling to and from the nonlocal mode, and ii) a local phase gradient with 100% efficiency is compatible with a vertically symmetric structure, while a nonlocal phase gradient with 100% efficiency is not.

As for the first point, in the traditional phenomenology of metasurfaces, a local phase gradient anomalously deflects light to the ±1st diffraction orders. The design and implementation of this functionality treats the meta-unit cell as black box: the near-field details are ignored, and only the net scattering phase is important. In contrast, a nonlocal phase gradient as implemented in refs. [60–64] anomalously deflects light to the ±2nd diffraction order. The discrepancy is due to the need to explicitly consider the near-field of the meta-unit, rather than the net scattering

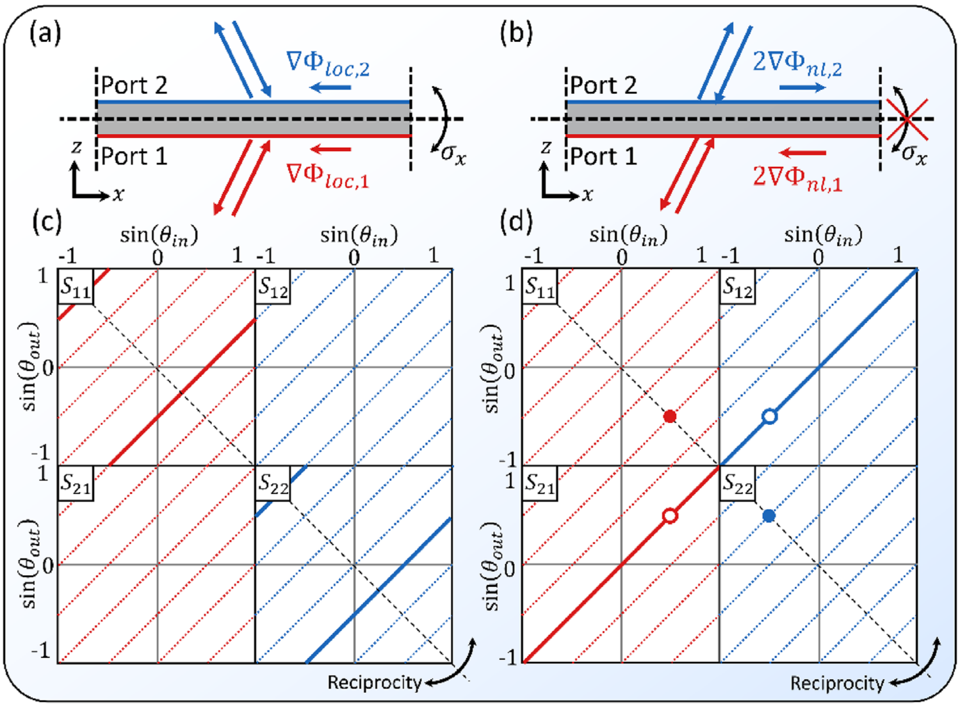

Figure 5. Comparison of ideal local and nonlocal phase gradients. a) Schematic showing how a conventional, local phase gradient provides a momentum kick that is equal in both magnitude and direction regardless of which side light is incident from; this is compatible with (though does not require) the reflection symmetry σ_x. b) The schematic scattering matrix for (a). c) Schematic showing a nonlocal phase gradient, providing a momentum kick equal but opposite depending on the direction of incidence. d) The schematic scattering matrix for (c).

phase: light couples into the metasurface with a coupling phase determined by the chosen perturbation, and then couples back to free-space with another factor of the same magnitude. Given the correlations across the in-plane direction, we must tailor the coupling phase to vary across 2π to avoid a discontinuity in the near-field that would cause destructive interference in the nonlocal mode. A coupling phase gradient that varies over 2π confers a total of 4π upon coupling back out to free-space. In this context, we note that since no metasurface is truly local, such near-field destructive interference may also impact the performance of conventional local metasurfaces, and that future efforts in improving anomalous deflection efficiencies may aim to eliminate such near-field phase discontinuities.

The second point clarifies that the local phase gradient is compatible with a symmetric response when incident from one side (port 1) of the metasurface compared to the opposite side (port 2) (see **Figure** 5a). That is, a local phase gradient for light incident from port 1, $\nabla\Phi_{loc,1}$ may be identical to the local phase gradient for light incident from port 2, $\nabla\Phi_{loc,2}$. We note that the reflection response from each port may also be distinct, for instance in Janus metasurfaces,[132] but the conventional response is symmetric. In contrast, a nonlocal phase gradient is constrained by reciprocity and energy conservation to act differently when light is incident from port 1 versus port 2. At the band-edge frequency, a nonlocal phase gradient shifts the band-edge momentum for excitation from port 1 by $\nabla\Phi_{nl,1}$, retroreflecting light incident at that momentum for a net momentum transfer of $2\nabla\Phi_{nl,1}$. However, considering light coming from port 2, light incident at $k_{in} = -\nabla\Phi_{nl,1}$ must also engage with the resonant response: transmitting without distortion would violate reciprocity. Instead, the light from port 2 is retroreflected when incident with $\nabla\Phi_{nl,2} = -\nabla\Phi_{nl,1}$ (see Figure 5b). This response has 180° rotation symmetry about the center of the device, while the local response has reflection symmetry. This behavior is consistent with the interpretation that the nonlocal phase gradient "tilts" the decay profile of the q-BIC from normal incidence to both ports, rightward to port 2 and leftward to port 1 and, hence, the q-BIC only engages light from the corresponding plane waves.[62] We note that the eigen-waves depicted in Figure 4b,d,f are implementable by controlling the magnitude and orientation of the "tilt" angle depicted in Figure 5b: wrapping it azimuthally encodes OAM and changing its magnitude radially encodes a focusing (defocusing) effect.

Figure 5c,d show schematic scattering matrices of the processes for Figure 5a,b, a scalar, two-port form of the vectorial, one-port schematics used in ref. [64]. The scattering matrix here is of the form $S_{ij}(k_{in}, k_{out})$, containing the complex amplitude of scattering for light incident with momentum k_{in} from port j to light outgoing with momentum k_{out} to port i. In a phase gradient device, conservation of momentum precludes coupling except when $k_{out} = k_{in} + mk_G$, where m is the diffraction order and k_G is the momentum kick provided by the grating. In the schematics, these contours are marked by red lines (for light incident from port 1) and blue lines (for light incident from port 2); the white area corresponds to zero values due to momentum conservation, that is, these are sparse matrices with most entries being zero. The lines are dotted when the diffraction efficiency is nevertheless zero, and solid when the magnitude is unity. The four quadrants correspond to different combination of input and output ports, and each quadrant depicts the response as a function of input and output momentum. Arranged in this manner, reciprocity requires these schematics to be symmetric around the black dashed line along the diagonal, enforcing $S_{ij}(k_1, k_2) = S_{ji}(-k_2, -k_1)$.

These schematics depict each of the ways local and nonlocal phase gradient devices differ, additionally highlighting the angular selectivity of nonlocal phase gradients. In the ideal local case, the $m = -1$ diffraction order is unity over the entire range of incident angles over which it is supported (outside this range, i.e., $k_{in} < -k_0 + k_G$, light must reflect or transmit to other diffraction orders). In contrast, in the nonlocal case the vast majority of the response is simply specular transmission, punctuated by the sharp anomalous resonance. Here, the solid disks denote the resonantly reflected light, while the open disks denote the corresponding transmission dip. By conservation of energy, any vertical slice must pass through either one of each type of these disks, or none at all. By time-reversal symmetry, each horizontal line must behave in the same manner. Hence, a Fano resonance forms a "box" of events in these schematic matrices and, since the transmission dip is constrained to the solid line (the specular response that acts as the background continuum of states in the Fano resonance), the nonlocal phase gradient serves to slide the open disk of the q-BIC along that line.

We do not claim that the behavior shown in Figures 5c,d is the only possible behavior of a nonlocal device. Future work will explore novel functionalities, including cases where the anomalous diffraction is transmissive, and where both the local and nonlocal responses simultaneously impart distinct wavefront transformations. Furthermore, Figures 5b,d depict the band-edge response only, where this is a single q-BIC mode; at other frequencies along the band, there are two counterpropagating modes and, in the corresponding schematics, the disks split into two, in each case dividing equidistantly along the 45° line, and maintain the symmetry about the −45° line enforced by reciprocity.

4.2. Spatial and Spectral Selectivity

A key aspect of the response of nonlocal devices is their selectivity, which can be manifested as both spectral selectivity (a narrowband resonant response) and spatial selectivity (which corresponds to angular selectivity in the case of an infinite periodic device). A selective metasurface only engages with light over a narrow range of incident conditions, broadly construed to include wavelength, wavefront and polarization (governed by the selection rules). The selectivity is strongly associated with nonlocality, where highly local devices have a low degree of selectivity and highly nonlocal devices have a high degree of selectivity. Here we stress that these two selectivities are independently tunable by tuning the optical lifetime and band structure of the q-BIC, and mention a few examples of each category.

To guide our discussion, we consider a mode following the parabolic dispersion $\omega_r(k) = \omega_0 + bk^2/2$, where the resonant frequency ω_r at the band-edge is ω_0 and the Taylor expansion coefficient b is a measure of the curvature of the band structure. The mode additionally has a radiative lifetime $\tau_r = Q/\omega_r$, independent of the incident angle (i.e., this is an artificial *q-BIC*).

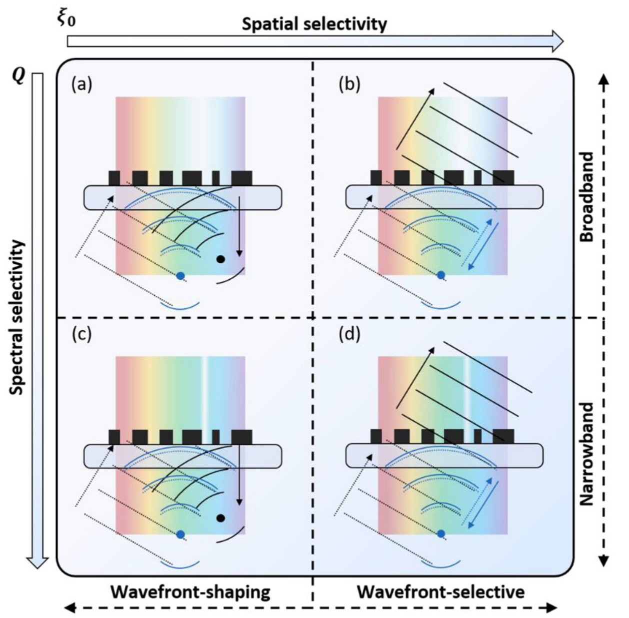

Figure 6. Spatial and spectral selectivity. A reflective wavefront transformation in a device with low spectral selectivity a,b) is broadband, while in a device with high spectral selectivity c,d) it is narrowband. A device with low spatial selectivity (a,c) anomalously reflects a wide range of incident waves, while a device with high spatial selectivity (b,d) only anomalously reflect a narrow range of incident waves. The incident waves are marked by dashed lines, and the scattered waves by solid lines. Diffractive nonlocal metasurfaces fall primarily into (c,d), where they can be further categorized as wavefront-shaping (c) or wavefront-selective (d).

As shown in,[64] the characteristic length travelled in plane by the band-edge mode is $\xi_0 = \sqrt{b\tau_r}$, which we call the nonlocality length. This length determines the extent of lateral correlations of the optical response; roughly, a device is local if $\xi_0 < \lambda$ and is nonlocal if $\xi_0 > \lambda$. While in general the selectivity grows proportionally to ξ_0, the constituent parameters b and Q distinctly affect the spatial and spectral selectivities.

Figure 6 depicts four exemplary device responses in the space spanned by the spatial and spectral selectivity of the device. Figure 6a shows a highly local device, reflecting, and focusing a wavefront over a broad range of wavelength and incident wavefronts. For instance, both a plane wave off normal incidence (black dashed lines) and a point source at twice the focal distance (blue dashed lines) are efficiently anomalously reflected. This functionality is typical of reflective metasurfaces.[133,134] In contrast, Figure 6b shows a device in which one of the wavefronts is reflected over a broad bandwidth, but the other is transmitted. While rarer, such broadband angularly selective devices have been demonstrated, including plasmonic Brewster devices,[135] broadband angular selective dielectric stacks,[136] and broadband angularly selective emitters.[137] Next, Figure 6c depicts a narrowband nonlocal metasurface anomalously reflecting a wide range of incident light, while Figure 6d depicts a narrowband nonlocal metasurface anomalously reflecting only one of the incident wavefronts. The former case is a wavefront-shaping nonlocal metasurface acting spatially like a highly local metasurface in that it efficiently applies a designed wavefront transformation over a wide range of incident wavefronts, but it does so

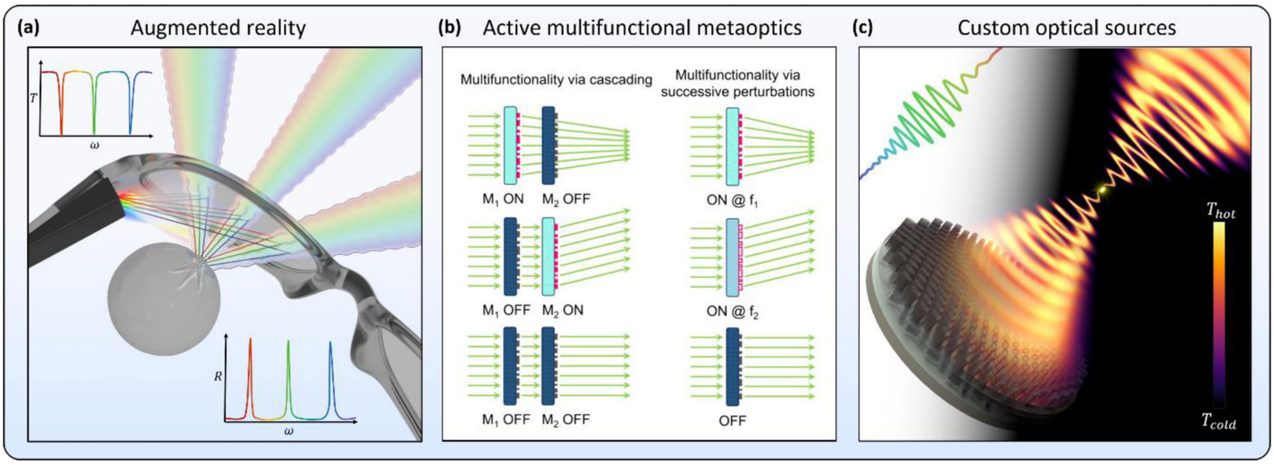

Figure 7. Applications of diffractive nonlocal metasurfaces. a) The exclusive wavefront-shaping response at resonance may enable ideal beam combiners for augmented reality applications with full field-of-view operation. b) The enhancement of light-matter interactions inherent to the high-Q metasurfaces, combined with wavefront shaping properties, promise high efficiency spatial modulators. Reproduced with permission.[150] Copyright 2021, De Gruyter. c) A wavefront-selective nonlocal metasurface populated in-plane by optical loss, gain, or waveguide, promises truly compact custom optical sources. Reproduced with permission.[64] Copyright 2021, American Physical Society.

exclusively within a narrow bandwidth. This functionality was demonstrated in refs. [61, 65] Such a device could be useful, for instance, in wavelength-exclusive imaging, where plane waves at a range of angles must be focused to distinct points. In contrast, the latter case is a wavefront-selective nonlocal metasurface only fully resonating for a narrow range of incident conditions, as demonstrated and discussed in ref. [63] Such devices could be useful, for instance, when only a narrow range incoming beams should be engaged, such as in thermal metasurfaces.[64]

5. Future Opportunities

The novel behavior of nonlocal metasurfaces promises a range of future opportunities. First, the combination of narrowband wavefront shaping and broadband transparency suggest opportunities for a multifunctional beam splitter of ideal use in augmented reality applications, and highly multifunctional cascaded meta-optics more generally. Second, the combination of enhanced light-matter interactions (high-Q operation) and wavefront manipulation portends breakthroughs in active, nonlinear and quantum photonic devices. Third, the combination of spatial correlations across the plane of the device and control over the optical scattering by locally manipulating the selection rules opens avenues for novel compact optical sources wherein the q-BIC is populated in-plane by loss, gain, or a waveguide. We briefly discuss these opportunities, seen in **Figure 7**, in the following.

5.1. Highly Multifunctional Meta-Optics for Augmented Reality

The selectivity of a nonlocal optical response is particularly well suited for adding multifunctionality to a compound meta-optic. Since the desired optical response is imparted exclusively at resonance, the broadband response may simply be transparent. This suggests that multiple diffractive nonlocal metasurfaces may be cascaded,[65] each targeting a narrow bandwidth of operation. This concept is compatible with the intrinsic multifunctionality of a single metasurface rooted in the orthogonality of distinct perturbations,[61] meaning each cascaded element may target a discrete set of wavelengths rather than just one.[65] This is to say, a compound meta-optic composed of cascaded nonlocal metasurfaces is capable of an unprecedented degree of multispectral control of light, where each functionality is rationally designable independently of the rest to first order in perturbation theory. This orthogonality in design is due to the origins of the working mechanism in symmetry protection; it is therefore a generic feature of these systems.

We suggest in particular augmented reality as an application uniquely benefiting from this combination of features: the broadband transparency of such a metasurface (or compound meta-optic) may let the majority of external information reach the user, while the narrowband anomalous reflection may redirect artificially generated information to the user. The use of metasurfaces for augmented reality applications has received significant recent attention[138–143] owing to the light-weight, compact footprint and ability to serve as a focusing element for magnification in the imaging system. However, broadband metasurfaces are inherently limited by reciprocity: the efficiency with which artificial information is focused is the same as the efficiency with which external information is distorted or lost. In contrast, we envision in Figure 7a, a nonlocal meta-optic serving as an ideal beam splitter, rejecting from the external world information exclusively within three narrow red, green, and blue channels, while reflecting and focusing information encoded in laser light. This exclusive focusing of narrow wavelength channels is a unique prospect born of the selectivity inherent to nonlocality.

5.2. Active, Nonlinear, and Quantum Metasurfaces

Among the many advantages of diffractive nonlocal metasurfaces, the supported states have arbitrarily long lifetimes, conferring greatly enhanced light-matter interactions by construction. At the same time, the remaining degrees of freedom can introduce additional functionalities. This platform is therefore highly promising for combining metasurface physics with active,[144–146] nonlinear,[147,148] and quantum technologies.[11,149]

For active metasurfaces, the long optical lifetime both enhances the impact of a small change in refractive index, as well as reduces the spectral distance the resonant wavelength needs to shift to achieve full modulation. We note in particular past demonstrations of artificial q-BIC concepts combined with electro-optic modulation[80,111] and thermo-optic modulation,[150] implying that combining these approaches with perturbations that encode wavefront transformations[150] is the readily obtainable next step to make an impact in the framework of active metasurfaces. Furthermore, these concepts offer a platform for advanced switchable surfaces[144] that provide more than a binary choice of states (seen in Figure 7b). Finally, two spectrally overlapping Fano resonances born of modes with opposite decay symmetries[115] may harness the highly transmissive phase manipulation concepts of Huygens' metasurfaces[151] in combination with active materials;[152] when implemented in a diffractive nonlocal metasurfaces, highly efficient transmission type active metasurfaces may be possible.

For nonlinear and quantum metasurfaces, the electric-field enhancement from long optical lifetime increases the efficiency of harmonic processes. While this has been achieved in plasmonic metasurfaces,[153,154] the loss limits the potential of these approaches compared to low-loss dielectric materials. In order to recover the large electric field enhancement despite removing the metallic materials, recent efforts have shown the high-Q performance of artificial q-BICs may be used.[77,78] This suggests that nonlinear metasurfaces with wavefront-shaping behaviors[155,156] may be implemented in tandem with the unbounded Q-factors of q-BICs. Similar considerations suggest the applicability of this novel platform for advancing quantum metasurfaces,[157,158] where the nonlocality of the classical aspects of the system may provide further tailoring of the quantum aspects.[154]

5.3. Leaky-Wave Metasurfaces

Finally, the leaky-wave nature of nonlocal metasurfaces may be leveraged directly as the q-BIC is populated from within the plane rather than from external sources. The perturbative control of the leaked portion holds irrespective of the illumination, though the Fano interference is absent without the direct optical pathways (Fabry–Perot resonance). Hence, diffractive nonlocal metasurfaces are a platform capable of custom emission, as recently demonstrated for thermal emission in ref. [64]. These concepts will guide future experimental work establishing a new paradigm of compact optical sources (Figure 7c) that achieves a long-term goal of flat optics: complete generation of a custom wavefront within a single thin film. Past thermal radiation, we anticipate compatibility with luminescence,[130,131] fluorescence,[159] and even lasing.[160,161]

Alternatively, if the q-BIC is fed laterally from an in-plane waveguide, a q-BIC metasurface promises advancing and generalizing holographic grating couplers[162–166] or leaky-wave antennas.[167–169] In the absence of a perturbation, the structure is a subwavelength waveguide grating;[170] but in the presence of a period doubling perturbation, a portion is leaked to free space. Judicious design of the perturbation may tune the amplitude, phase, and polarization of the leaked state.[171] Spatially varying these perturbations holds the promise of a coupler for custom vector beams or even holograms, implying that q-BICs are directly applicable to the broad integrated photonics platform.

6. Conclusions

In this work, we have provided a perspective on metasurfaces able to manipulate both spatial and spectral degrees of freedom, enabled by advances in symmetry-driven control of long-lived resonant states. The origin of these phenomena lies in symmetry, implying that geometric design knobs may tune distinct optical responses with independence, advancing the capabilities of flat optics. Future work incorporating advanced optimization techniques into these physics-driven design approaches may open even further avenues for enhanced performance. In addition to extending the degree of multifunctionality compared to past efforts, this new class of devices offers novel directions for multifunctional operation. Distinct from conventional wavefront-shaping metasurfaces, they may manipulate arbitrarily narrow bandwidths, while the (non-resonant) remainder of the optical spectrum has a conventional specular response. Distinct from conventional gratings and photonic crystal slabs, they may spatially manipulate the scattering of the guided modes. In addition, distinct from most flat optical devices (including conventional local phase gradients), the responses are asymmetric with respect to the direction of incidence, and may exclusively engage an arbitrary polarization of choice (including pure circular dichroism).

These diffractive nonlocal metasurfaces are a generalized class of flat optical devices explicitly leveraging nearest-neighbor interactions that confer controllable selectivity to the optical response, while nevertheless maintaining robust control over the local wavefront shape. The spatial and spectral selectivity are simultaneously tunable by the band structure and perturbation strength, suggesting a wide range of behaviors and applications. When highly spectrally selective, the enhanced light-matter interactions inherited from the long optical lifetimes confer many advantages for the efficiency of active, nonlinear, and quantum photonics, while simultaneously enabling the incorporation of wavefront and polarization manipulation. When highly spatially selective, a limited range of incident waves efficiently engage the strong near-fields, conferring an exclusivity in the far-field response of great interest for augmented reality applications and devices acting as custom optical sources, including thermal metasurfaces and metasurface lasers. When not spatially selective, the long-lived modes are laterally localized, enabling excitation with a high numerical aperture, advantageous for nonlinear, active, and frequency-selective optics. Especially in combination with advanced computational techniques, we believe that diffractive nonlocal metasurfaces represent a next-generational platform for highly efficient and novel multifunctional flat optics in which entire optical systems (even including sources) are compactified into structured thin films.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

bound states in the continuum, fano resonance, metasurfaces, nonlocality, nonlocal metasurfaces, wavefront shaping

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